

Perception as a Construction: A Case Study in Sensory Neuroscience

By now you have learned that our brains do not passively receive the world and represent each pixel exactly as a camera would. Instead, the nervous system actively constructs perceptions based on its expectations of what it thinks it should see. No where is this clearer to see than in conditions where sensory processing diverges from the norm, revealing the hidden mechanisms the brain uses to build our experience of reality. In this assignment, you will explore one such condition and use it as a window to understand how sensory systems work.

Instructions

1. Choose a condition: Select one sensory condition from the list below or choose your own (check with me first if you're unsure whether your choice is a good fit).

Example Conditions:

- Synesthesia (e.g. associating colors with numbers, sounds etc.)
- Phantom limb pain
- Prosopagnosia (face blindness)
- Congenital insensitivity to pain
- Auditory hallucinations
- Motion blindness (akinetopsia)
- Taste or smell blindness (ageusia/anosmia)
- Tinnitus
- Alice in Wonderland syndrome (Todd's syndrome)
- Visual neglect (hemispatial neglect)
- Sensory dysfunction in disorders like Autism Spectrum Disorders, Schizophrenia, Alzheimer's Disease, etc.

2. Write a 400-500 word response that does the following:

- a. Neuroscience Explanation: Explain what is happening in the brain in this condition. What system or pathway is affected? What mechanisms are impacted or work differently than baseline conditions? Be as specific as you can.
- b. Conceptual Exploration: What does this condition reveal about how the healthy brain constructs perceptions or integrates information (across senses, time, or brain regions)? Does this condition challenge any intuitions you had about perception being a direct, faithful readout of the world? Are there aspects of normal perception that we take for granted that this condition makes visible?

3. Cite at least one source beyond the course materials (textbook, lectures). You could use a scientific article from a reputable magazine or website (Quanta, Scientific American etc.), a peer-reviewed publication, or a book. Any citation format is fine.

A Note on AI Use: Using LLMs to aid you in your research, help you find sources, and explore what is known or not known about a topic is a wonderful way to use an LLM for this assignment. However, you should not use the LLM to write text that you copy, paste, and submit, even if you spend time editing the text. Your written reflections should be in your own words and of your own effort. If you have questions about whether use of AI crosses the boundaries of academic integrity for this (or any) assignment, please feel free to reach out to me and to re-visit the course syllabus.

Rubric (5 points total)

2 points: Neuroscience explanation is an accurate, specific account of what's happening in the brain

2 points: Conceptual insight clearly articulates what the condition reveals more broadly about perception

1 point: A source is cited, word count is met